

ECE253 / CSE208 — Homework 1

Probability Theory

1. Let X and Y be discrete random variables with joint probability mass function

$$P_{X,Y}(x,y) = \begin{cases} c(x+y), & x \in \{0,1,2\}, y \in \{0,1\}, \\ 0, & \text{otherwise,} \end{cases}$$

where $c > 0$ is a normalization constant.

- (a) Determine the value of c .
 - (b) Compute the marginal PMFs $P_X(x)$ and $P_Y(y)$.
 - (c) Compute the conditional PMF $P_{X|Y}(x | Y = 1)$.
 - (d) Are X and Y independent? Justify your answer mathematically.
2. A communication link transmits packets in discrete time slots. Each packet transmission succeeds independently with probability $p \in (0,1)$. Let $Z_i \in \{0,1\}$ denote the outcome of the i -th transmission, where $Z_i = 1$ indicates success.

- (a) Identify the distribution and probability mass function of Z_i .
- (b) Let $S_n = \sum_{i=1}^n Z_i$ denote the total number of successful transmissions in n slots.
 - (i) Write down the PMF of S_n .
 - (ii) Compute $P(S_n \geq k)$ for a given integer k , expressing your answer as a finite sum.
- (c) Suppose it is observed that at least one packet was successfully transmitted, i.e., $S_n \geq 1$. Compute

$$P(Z_1 = 1 | S_n \geq 1),$$

and explain why this conditional probability differs from $P(Z_1 = 1)$.

- (d) Let T denote the time index of the first successful transmission. Compute $P(T > m | S_n \geq 1)$ for $m < n$, and provide an intuitive interpretation.
3. A device repeatedly attempts to transmit a message until the first successful transmission occurs. Each attempt succeeds independently with probability p .
- (a) Define a random variable T representing the number of attempts until the first success. Identify its distribution and write its PMF.
 - (b) Compute $P(T > k)$ for $k \geq 0$.
 - (c) Compute $\mathbb{E}[T]$.
 - (d) Show that

$$P(T > m + n | T > m) = P(T > n),$$

and briefly explain the modeling significance of this property.

4. Consider a generalized Monty Hall problem with 100 doors. One door hides a prize and the remaining 99 doors hide goats. You initially select one door. The host, who knows the prize location and never opens the prize door, then opens 98 doors, all of which reveal goats. You are then offered the option to stay with your original choice or switch to the only remaining unopened door.
- (a) Compute the probability of winning the prize if you stay with your original choice.
 - (b) Compute the probability of winning the prize if you switch.
 - (c) Suppose instead that the host opens only k goat doors, where $1 \leq k \leq 98$. Derive the probability of winning if you switch as a function of k .
 - (d) Is switching advantageous for all k ? Justify your answer mathematically.
5. Assume that each person's birth month is independently and uniformly distributed over the 12 months of the year.
- (a) In a group of n people, derive an expression for the probability that no two people share the same birth month.
 - (b) Write the probability that at least two people share the same birth month.
 - (c) Find the smallest integer n such that the probability that at least two people share a birth month exceeds 0.5.
 - (d) For a fixed n , derive an exact or approximate expression for the probability that at least three people share the same birth month.